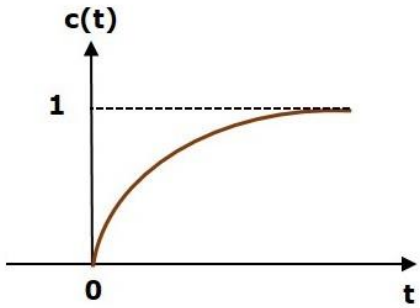
UNIT 4:

ROOT LOCUS TECHNIQUE AND BODE PLOTS

4.1 Concepts of Stability

A system is said to be stable, if its output is under control. Otherwise, it is said to be unstable. A **stable system** produces a bounded output for a given bounded input.

The following figure shows the response of a stable system.



This is the response of first order control system for unit step input. This response has the values between 0 and 1. So, it is bounded output. We know that the unit step signal has the value of one for all positive values of tincluding zero. So, it is bounded input. Therefore, the first order control system is stable since both the input and the output are bounded.

4.2 Types of Stability:

We can classify the systems based on stability as follows.

- Absolutely stable system
- Conditionally stable system
- Marginally stable system

4.2.1 Absolutely Stable System

If the system is stable for all the range of system component values, then it is known as the absolutely stable system. The open loop control system is absolutely stable if all the poles of the open loop transfer function present in left half of ‘s’ plane. Similarly, the closed loop control system is absolutely stable if all the poles of the closed loop transfer function present in the left half of the ‘s’ plane.

4.2.2 Conditionally Stable System

If the system is stable for a certain range of system component values, then it is known as conditionally stable system.

4.2.3 Marginally Stable System

If the system is stable by producing an output signal with constant amplitude and constant frequency of oscillations for bounded input, then it is known as marginally stable system. The open loop control system is marginally stable if any two poles of the open loop transfer function is present on the imaginary axis. Similarly, the closed loop control system is marginally stable if any two poles of the closed loop transfer function is present on the imaginary axis.

4.3 Asymptotic Stability

A time-invariant **system** is **asymptotically stable** if all the eigenvalues of the**system** matrix A have negative real parts. If a **system** is **asymptotically stable**, it is also BIBO **stable**. ... A **system** is defined to be exponentially **stable** if the **system** response decays exponentially towards zero as time approaches infinity.

A system is defined to be **uniformly asymptotically stable** if the system is asymptotically stable for all values of t_0.

A system is defined to be **exponentially stable** if the system response decays exponentially towards zero as time approaches infinity.

Asymptotic stability. ... Convergence requires a stronger notion called **asymptotic stability**. A point is an **asymptotically stable** equilibrium point of if: It is a Lyapunov**stable** equilibrium point of . There exists some open neighborhood of such that, for any , converges[15.2] to as approaches infinity.

Introduction to Root Locus:

Root locus is a graphical method for determining the roots of the characteristic equation of a system, when one of its parameters is varied through all possible values.

It is necessary to understand that the location of roots of the characteristic equation in the s-plane determines the stability and transient performance of the system. For the system to be stable all the roots of the characteristic equation must lie in the left

half of s-plane. The location of poles also determines the transient performance characteristics such as maximum overshoot, rise time and settling time. Therefore in order to understand the system performance completely it is necessary to determine the roots of the characteristic equation which is a polynomial in s (for n^{th} order system degree of polynomial is n). But determination of roots of a polynomial of degree greater than 3 is difficult.

Also in control systems one of the parameters (usually gain K) is varied to obtain the desired response. Hence it is necessary to know the roots of the characteristic equation for different values of gain. For this purpose the characteristic equation needs to be solved repeatedly with different values of K which is a tedious process. Hence there is a need for a method which readily gives the roots of the characteristic equation for all possible values of gain.

Root locus is the loci of the roots of the characteristic equation as one of the parameters of the system is varied from 0 to ∞. After constructing the root locus, the roots of the characteristic equation are determined for any given value of the variable parameter.

4.4 ROOT LOCUS CONCEPT:

Consider a second order system shown below

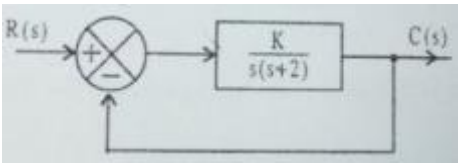


Fig 4.1 Second order System.

The open loop transfer function of the system is $G(s) = \frac{k}{s(s+2)}$. The characteristic equation for the system is

$$1 + G(s)H(s) = \frac{k}{s(s+2)} = 0$$

i.e., $s^2+2s+k=0$

The roots of characteristic equation (or poles of the closed loop system) are given by

$$s_1 = -1 + \sqrt{1-k}\ ,$$
$$s_2 = -1 - \sqrt{1-k}$$

The location of roots in the s plane for different values of K are as follows

(i) When K = 0, the roots are $s_1 = 0$ and $s_2 = -2$

The open loop poles of the system are also at s = 0 and s= -2.

When K = 0, the roots of the characteristics equation are at open loop poles.

(ii) When 0<K< 1, the roots are given by $s_1 = -1 + \sqrt{1-k}$ and $s_2 = -1 - \sqrt{1-k}$

The roots are real and distinct. This condition corresponds to overdamping of the system.

Both the roots in this case lie on the real axis.

(iii) When K = 1 the roots are real and equal.

$$S_1 = S_2 = -1$$

This condition corresponds to critical damping of the system.

(iv) When 1 < K < ∞, the roots are complex.

$$s_1 = -1 + j\sqrt{k-1} \text{ and } s_2 = -1 - j\sqrt{k-1}$$

In this case the real part of the poles is equal to -l for all values of K greater than 1. But the imaginary part increases from 0 to ∞. This condition corresponds to underdamping of the system. The locus of roots of characteristic equation when K is varied from 0 to ∞ is shown in Fig.4.2. This is the root locus of the system shown in Fig.4.1

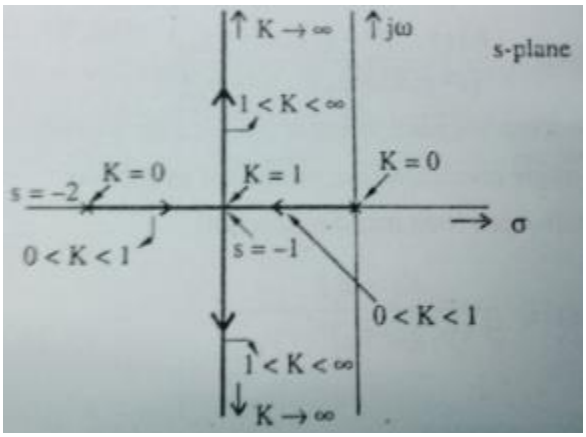


Fig. 4.2 Root Locus of $s^2+2s+k=0$ as a function of k

The following observation can be made from the above fig4.2.

(a) There are two root loci starting at open loop poles S=0 and S=-2 (for K= 0).
(b) As K increases, the root locus starting at S=0 moves towards left and one starting at S=-2 moves towards right.
(c) When K=1, the two root loci meet at s=-1 and breakaway from the real axis at this point as increases.
(d) When K > 1, one root locus moves upwards and other downwards. The real part remains constant.

The method shown here for drawing the root locus is for a second order system. But for higher order systems, it is difficult to obtain a direct solution for the roots. Hence a method developed by ***W.R. Evans*** is used for drawing the root locus.

4.5 DEFINITION OF ROOT LOCUS DIAGRAM:

The characteristic equation of a system is given by,

$$1+G(S)H(S)=0 \qquad ..(4.1)$$

$$\text{Or} \quad G(S)H(S)=-1 \qquad ..(4.2)$$

Since G(s) H(s) is a complex quantity, for satisfying Equation (4.2) the following conditions have to be satisfied

Angle condition:

$$\angle G(S)H(S) = \pm(2k+1)180° \qquad (4.3)$$

Where k=0,1,2,3,….

Magnitude Condition:

$$|G(S)H(S)| = 1 \qquad (4.4)$$

If a point on the s-plane is a point on the root locus of the system, then it has to satisfy the conditions given in Equation (4.3) and Equation (4.4).

In general open loop transfer function G(s) H(s) of a system is of the form:

$$G(S)H(S) = \frac{k(s+z_1)(s+z_2)\ldots\ldots\ldots(s+z_m)}{(s+p_1)(s+p_2)\ldots\ldots\ldots(s+p_n)} \qquad (4.5)$$

For drawing the root locus the location of zeros and poles of G(s) H(s) has to be known.

The magnitude and angle conditions are explained as follows,

Consider a system with open loop transfer function,

$$G(S)H(S) = \frac{k(s+z_1)}{(s+p_1)(s+p_2)(s+p_3)}$$

Where $-Z_1$ is a zero and $-P_1, -P_2, -P_3$ are poles of G(S)H(S). They are plotted in the S-plane as shown in fig 4.3

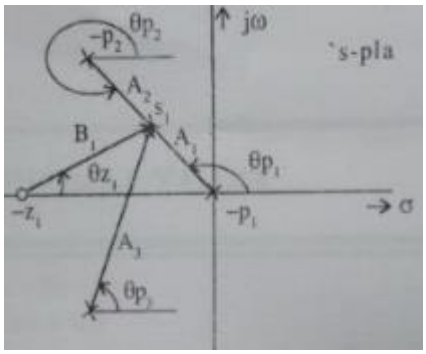


Fig. 4.3 Pole-Zero plot of the system

If S_1 is a point on the root locus, then it has to satisfy (4.3) and (4.4)

Angle condition:

$$\angle G(S)H(S) = (\angle s_1 + z_1) - (\angle s_1 + p_1 + \angle s_1 + p_2 + \angle s_1 + p_3) = \pm(2k+1)180°$$

Here, (s_1+z_1) is the vector drawn from $-z_1$ to s_1.

$\angle(s_1 + z_1)$ represents the argument of this vector.

$$\therefore \angle G(s_1)H(s_1) = \left[(\theta_{z1}) - (\theta_{p1} + \theta_{p2} + \theta_{p3})\right] = \pm(2k+1)180°$$

Magnitude Condition:

$$|G(S)H(S)| = 1$$

$$\frac{|k||(s+z_1)|}{|(s+p_1)||(s+p_2)||(s+p_3)|} = \frac{kB_1}{A_1A_2A_3} = 1$$

Where, B_1 is the length of the vector $(s+z_1)$.

A_1, A_2, A_3 are the length of the vectors $(s+P_1)$, $(s+P_2)$ and $(s+P_3)$ respectively.

The magnitude condition is used to determine the value of K at a point S_1 on the root locus.

$$k = \frac{(Product\ of\ lenghts\ of\ vectors\ drawn\ from\ open\ loop\ poles\ to\ the\ point\ S_1)}{(Product\ of\ lengths\ of\ vectors\ drawn\ from\ open\ loop\ zeros\ to\ the\ point\ S_1)}$$

4.6 STEPS TO DRAW THE ROOT LOCUS DIAGRAM/ CONSTRUCTION RULES:

Rule 1: ***Starting points:*** The root locus starts (K = 0) at open loop poles [i.e., G(s)H(s)].

Rule 2: ***Ending points:*** The root locus ends(K =∞) at open 1oop Zeros 0r at infinity.

Rule 3: ***Number of separate root loci:***

Let, P=Number of finite poles of G(S) H(S)

Z =Number of finite zeros of G(s) H(S)

N =Number of separate root loci.

Then, N=P (if P > Z)

N=Z (if Z > P)

The three rules are explained in example 4.1

Example 4.1: Consider a system with open loop transfer function

$$\boldsymbol{G(S)H(S) = \frac{k(s+2)}{s(s+1)(s+4)}}$$

Sol.

Rule 1:- The root loci starts at s = 0, s = -1, s = -4.

Rule 2:- One root locus ends at s = -2.

The other two root loci terminate at ∞.

Rule 3:- Number of open loop poles, P = 3

Number of open loop zeros, Z = 1

Here, P > Z

∴Number of separate root loci, N = P = 3.

From the magnitude condition,

$$|G(S)H(S)| = 1$$

$$\frac{|s+2|}{|s||(s+1)||(s+4)|} = \frac{1}{|K|} \ldots\ldots\ldots\ldots (*)$$

As K →0, Equation (*) approaches infinity and hence root locus approaches the poles, S=0, -l and -4. Because, the left hand side of Equation (*) approaches infinity when s approaches the poles.

As K→∞, Equation (*) approaches zero in value and root locus approaches the zero S=-2/

Here 3 root loci are starting at the open loop poles and one root locus is terminating at open loop zero. The other two root loci terminate at infinity

NOTE: If P=Z, no root loci terminate at infinity

Rule 4: ***Asymptotes of root loci:***

For higher values of (K→∞), the root loci approaches zeros at infinity along asymptotes,

The number of asymptotes is equal to P-Z. The angle of asymptotes with respect to positive real axis is given by,

$$\theta_k = \frac{(2k+1)}{P-Z}180°$$

Where, K=0,1,2,…..(P-Z-1)

Rule 5: ***Intersection of the asymptotes (Centroid):***

The asymptotes explained in rule 4 intersect on the real axis. The point of intersection of the asymptotes is given by,

$$\sigma_1 = \frac{[Sum\ of\ poles\ of\ G(S)H(S)] - [Sum\ of\ zeros\ of\ G(s)H(S)]}{P-Z} = \frac{\sum p - \sum z}{P-Z}$$

Explanation: In root locus of fig 4.2, two root loci are starting at S=0,-2. They are terminating at infinity along asymptotes. The angle of the asymptotes are given by,

$$\theta_k = \frac{(2k+1)}{P-Z}180°, \qquad k = 0,1, \dots \dots .. (P-Z-1)$$

$$= \frac{(2k+1)}{2-0}180°, \qquad k = 0,1$$

$$=90°,270°$$

The asymptotes are intersecting at σ_1

And

$$\sigma_1 = \frac{[Sum\ of\ poles\ of\ G(S)H(S)] - [Sum\ of\ zeros\ of\ G(s)H(S)]}{P-Z}$$

$$= \frac{(0-2)}{(2-0)} = -1$$

NOTE: Rule 4 & 5 are not to be used when P=Z

Rule 6: Root loci on the real axis:

On a given section of real axis the root locus will be formed if the total number of poles and zeros of G(S)H(S) to the right of the section is an odd number

Explanation: Consider the open loop transfer function of example 4.1

$$G(S)H(S) = \frac{k(s+2)}{s(s+1)(s+4)}$$

The pole-zero plot of the above system is shown in fig 4.4

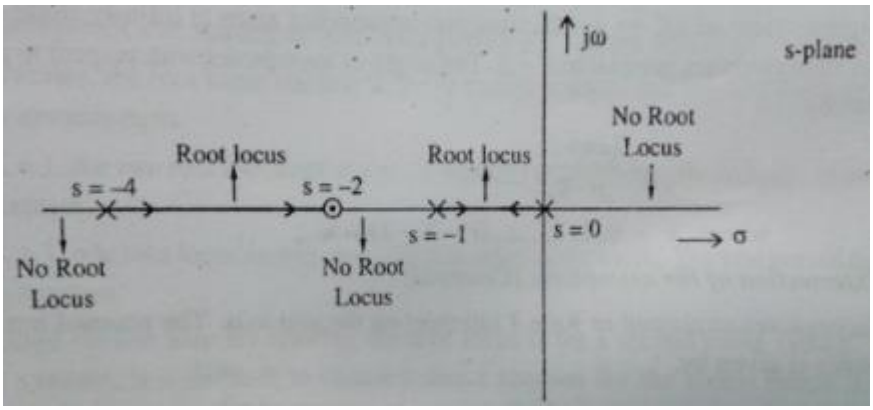


Fig 4.4 Pole-Zero plot

(a) Section between s=∞ and s = 0; In this section root locus is not found as the total number of poles and zeros to the right of the section is zero.
(b) Section between s =0 and s =-1; The total number of poles and zeros to the right of this section is 1. Hence root locus is found in this section.
(c) Section between s = -1 and s = 2; The total number of poles and zeros to the right of this section is 2 (even number). Hence root locus is not found in this section.
(d) Section between s = -2 and s=- 4; The total number of poles and zeros to the right of this section is 3 (odd number). Hence root locus is found in this section.
(e) Section between 5 = -4 and s =- The total number of poles and zeros to the right of this section is 4. Hence root locus is not found in this section.

Rule 7:- ***Points of intersection of root locus with imaginary axis:***

The points at which a root locus intersects the imaginary axis and corresponding value of K is found by applying Routh-Hurwitz criterion.

Rule 8:

(a) The angle of departure at a pole:

For a complex pole S_1 the angle at which root locus starts at this pole is given byθ_d,

where $$\theta_d = 180° + \angle G(S)H(S) \setminus s = s_1$$

When a simple pole is on the real axis, the angle at which root locus starts at this pole is 0° or 180°.

(b) The angle of arrival at a zero:

For a complex zero S_1 the angle at which root locus arrives at this zero is given by θ_A,

where $\theta_A = 180° - \llcorner G(S)H(S) \setminus s = s_1$

When a simple zero is on the real axis, the angle at which root locus terminates at this zero is 0° or 180°.

Rule 9*:- Breakaway points :*

Breakaway points are the points on the root locus at which multiple roots of the characteristic equation are found. At a breakaway point two or more root loci meet and breakaway as shown in Figure 4.5 (a), (b) and (c).

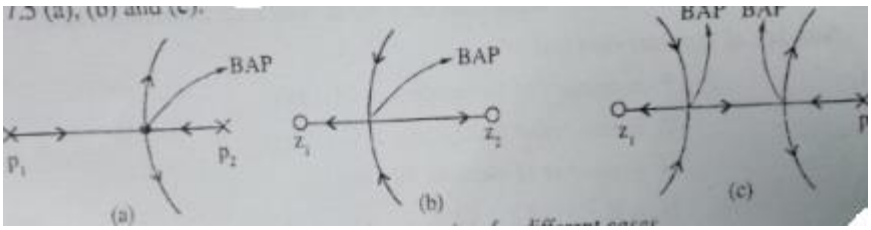


Fig 4.5 Breakaway points for different cases

In Fig. 5.5 (a), two root loci are starting at p_1 and p_2. These two root loci meet at Breakaway point and breakaway. In Fig. 5.5(b), two root loci which are starting at poles (not shown in fig), break in and terminate at the zeros Z_1 and Z_2. This point is also known as break in point.

In the first two cases two neighboring poles and neighboring zeros are present on the real axis. If root locus is present on the section between them (by Rule 6), then there is atleast one breakaway point in that section.

If root locus is present on the real axis between an open loop pole and a zero, there may not be any breakaway points in that section. However, if breakaway points are present between them their number will be two as shown in Fig. 4.5(c).

At a breakaway point more than two root loci can meet and breakaway. The root loci meet and leave a breakaway point at an angle of $\frac{180°}{n}$ apart, where n is the number of root loci meeting at the breakaway point.

Determination of breakaway points:

If $G(S)H(S) = \frac{KA(s)}{B(S)}$, then characteristic equation is written as

$$1 + G(s)\,H(s) = B(s) + KA(s) = 0$$

$$\therefore K = -\frac{B(S)}{A(S)}$$

On differentiating above Equation with respect to s and equating to zero,

$$\frac{dK}{ds} = -[B'(S) - A(s) - B(S)A'(S)] = 0$$

The roots of above Equation are breakaway points provided they are also points on the root locus.

Rule 10: ***Root loci are symmetrical with respect to real axis of s-plane***

Example **A unity feedback system has an open loop transfer function** $\boldsymbol{G(S) = \frac{K}{s(s^2+4s+13)}}$ **Sketch the root locus plot of the system**.

Sol. $$G(S) = \frac{K}{s(s^2+4s+13)} = \frac{k}{s(s+2-j3)(s+2+j3)}$$

l. Number of poles P = 3 at s=0,-2+j3,-2-j3.

2. Number of zeros Z = 0.

3. Number of separate root loci = N = 3 (P > Z).

4. Number of asymptotes = P - Z = 3.

Angle of asymptotes,

$$\theta_k = \frac{(2k+1)}{P-Z}180° \ldots\ldots\ldots k = 0,1,2$$

$$= \frac{(2k+1)}{3} 180°$$

$$= 60°, 180°, 300°$$

5. Asymptotes intersect on the real axis at σ_1

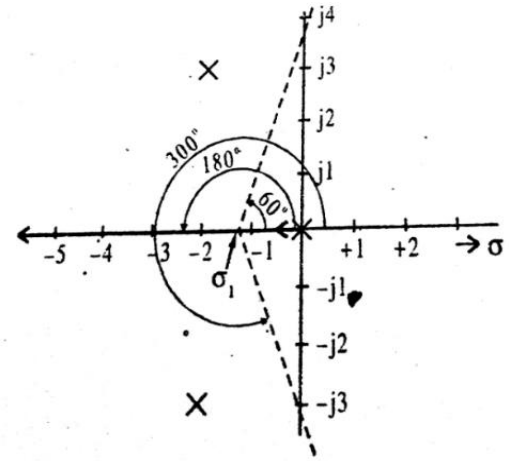
$$\sigma_1 = \frac{[Sum\ of\ poles\ of\ G(S)H(S)] - [Sum\ of\ zeros\ of\ G(s)H(S)]}{P - Z}$$

$$= \frac{(0 - 2 + j3 - 2 - j3)}{3}$$

=- l.333

6. Root locus on the real axis:

Draw the pole zero plot of the open loop system as shown in Figure below



By rule 6, the root locus is found on the real axis between s = 0 and s = -∞.

Asymptotes are drawn at angles of 60°, 180° and 300° at $\sigma_1 = -1.33$.

7. Angle of departure at complex poles.

Angle of departure at pole (-2 + j3), $\theta_{d1} = 180° + \llcorner G(S)H(S) \setminus s = -2 + j3$

$$\theta_{d1} = 180° + [(\llcorner k) - (\llcorner s + \llcorner s + 2 - j3 + \llcorner s + 2 + j3)]|_{s=-2+j3}$$

$$\theta_{d1} = -33.6°$$

Angle of departure at pole(-2-j3) is $\theta_{d2} = 33.6°$. The root locus leaves the complex pole (-2 + j3) at an angle of $-33.6°$ ($\theta_{d2} = -\theta_{d1}$ for complex conjugate roots)

8. Point of intersection of root locus with imaginary axis is determined by RH criterion: Characteristic equation is l + G(s) H(s) = 0

i.e., $1 + \frac{k}{s(s^2+4s+13)} = 0$

$s^3+4s^2+13s+K = 0$

Routh array:

S^3	1	13
S^2	4	k
S^1	(52-k)/4	
S^0	K	

When the root locus intersects the imaginary axis all the elements in a row are zero. This condition occurs when K=52. From the stability analysis it is found that when 0 <K < 52, root locus is in the left half of s-plane.

From s^2 row, Auxiliary equation is formed.

$A(S) = 4s^2+K=4s^2+52=0$

$$s = \pm j\sqrt{\frac{52}{4}} = \pm j3.6$$

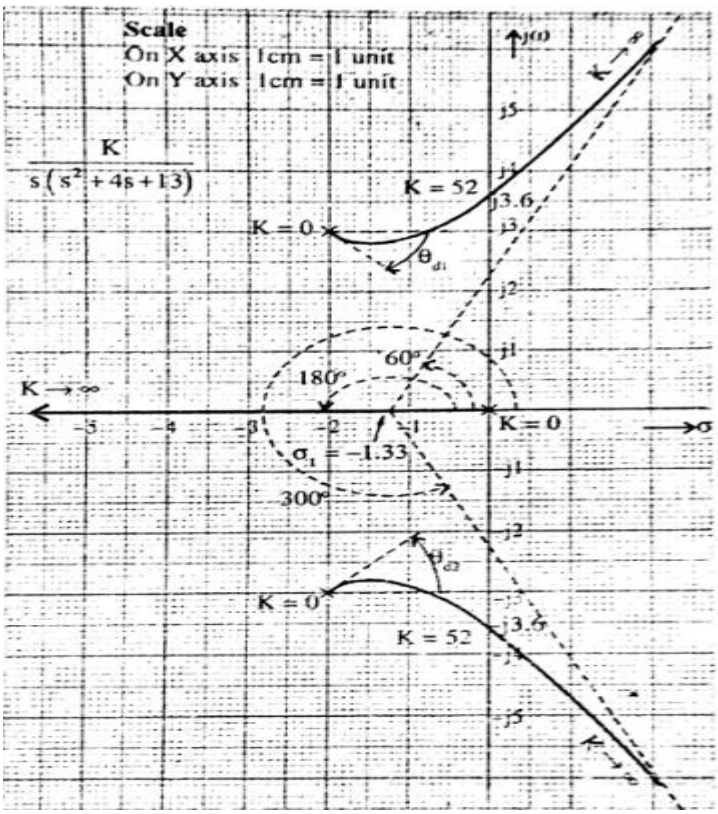
Root locus intersects the imaginary axis at $s = \pm j3.6$.

9. To determine breakaway points: $K = -(s^3 +4s^2+ 13s)$

$$\frac{dk}{ds} = -(3s^2 + 8s + 13) = 0$$

On solving this equation, s = - 1.33± j1.59. These are not points on the root locus. There are no breakaway points. By using the above information root locus is drawn as shown in Figure 4.7.

Note: In plotting root locus same scale has to be used for x and y axes.



Example : **Draw the Root Locus and determine the stability of the system with**

$$\mathbf{G}(\boldsymbol{S})\boldsymbol{H}(\boldsymbol{S}) = \frac{\boldsymbol{k}}{\boldsymbol{s}(\boldsymbol{s}+\mathbf{2})(\boldsymbol{s}+\mathbf{8})}$$

Sol. $$G(S)H(S) = \frac{k}{s(s+2)(s+8)}$$

l. Number of poles P = 3 at s=0,-2,-8.

2. Number of zeros Z = 0.

3. Number of separate root loci = N = 3 (P > Z).

4. Number of asymptotes = P - Z = 3.

Angle of asymptotes,

$$\theta_k = \frac{(2k+1)}{P-Z}180° \ldots\ldots\ldots k = 0,1,2$$

$$= \frac{(2k+1)}{3}180°$$

$$= 60°, 180°, 300°$$

5. Centroid $\sigma_1 = \frac{\sum p - \sum z}{P-Z}$

$$\sigma_1 = -3.33$$

6. Root Locus exists on Real axis between s =0 and -2 and beyond -8 upto -∞

7. To determine Break away point (BAP)

From the characteristic equation, obtain expression for K

Characteristic equation 1 + G(s) H(s) = 0

$$S(S+2)(S=8)+k=0$$

$$K=-[s^3+10s^2+16s]$$

$$S=-0.929, -5.736$$

Since s=-0.929 is a point on the root locus, it is a breakaway point.

S=-5.736 is not a point on the root locus. Hence it is not a breakaway point.

8. Crossing of the roots on imaginary axis

Routh array:

S^3	1	16
S^2	10	K
S^1	(16-0.1k)	
S^0	K	

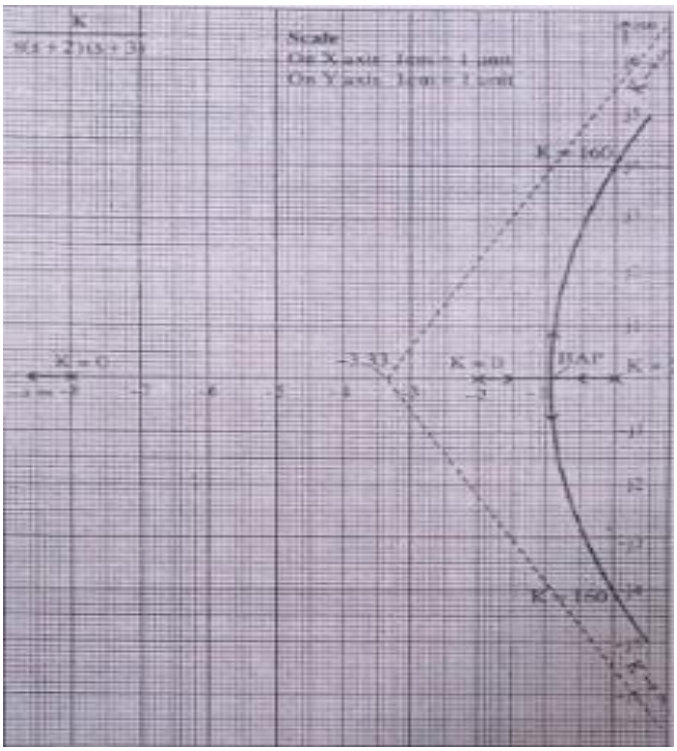
From s^1 row,16-0.1k=0

K=160

To determine the value of ω

$10s^2+k=0$

S=j ω thus $s = \pm j4$



Example : **A unity feedback system is defined by the characteristic equation**

$$s^3 + 6s^2 + 8s + K = 0$$

(i) Draw the root locus diagram for K varying from 0 to ∞

(ii) Find the range of K for which the system has damped oscillatory response.

(iii) Find the highest value of K which can be used before continuous oscillations occur.

(iv) What is the frequency of oscillations in (iii)

(v) Determine the value of K for a damping ratio of 0.8.

Sol. Characteristic eq. $s^3 + 6s^2 + 8s + K = 1 + G(S)H(S)$

$$= 1 + \frac{K}{s(s^2+6s+8)}$$

$$G(S)H(S) = \frac{K}{s(s+2)(s+4)}$$

l. Number of poles P = 3 at s=0,-2,-4.

2. Number of zeros Z = 0.

3. Number of separate root loci = N = 3 (P > Z).

4. Number of asymptotes = P - Z = 3.

Angle of asymptotes,

$$\theta_k = \frac{(2k+1)}{P-Z}180° \ldots\ldots\ldots k = 0,1,2$$

$$= \frac{(2k+1)}{3}180°$$

$$= 60°, 180°, 300°$$

5. Centroid $\sigma_1 = \frac{\sum p - \sum z}{P-Z}$

$$\sigma_1 = -2$$

6. Crossing of Root Locus on Imaginary axis

Characteristic eq. $=s^3 + 6s^2 + 8s + K$

From s^1 Row

$$\frac{48-k}{6} = 0 \quad or\ k = 48$$

The value of K when root locus crosses the imaginary axis is 48

Routh array:

S^3	1	8
S^2	6	K
S^1	(48-k)/6	
S^0	K	

From s^2 Row

$6s^2+K=6s^2+4800$

$S^2=-8$

And $\omega = 2.828$

7. To determine Break away point (BAP)

From the characteristic equation, obtain expression for K

Characteristic equation 1 + G(s) H(s) = 0

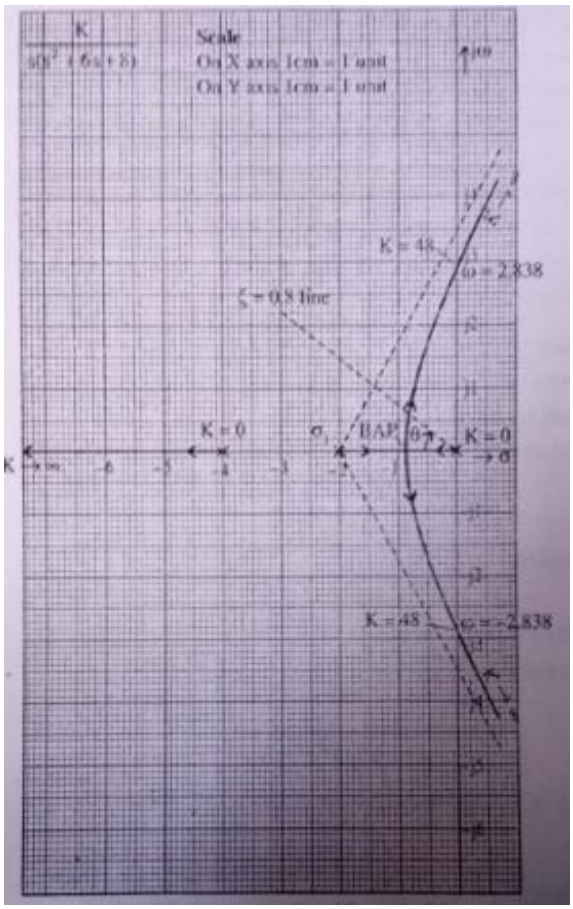
$$S(S+2)(S=4)+k=0$$

$$K=-[s^3+6s^2+8s]$$

$$S=-3.155 \text{ or } -0.845$$

The root locus exists on real axis in between 0and -2 and between -4 and -∞

Hence s=-0.845 is a breakaway point.



(ii) To determine the value of K for damped oscillatory response.

Determine the value of K at BAP.

Using the magnitude condition

$$|G(S)H(S)| = 1$$

$$\frac{|k|}{|s||(s+2)||(s+4)|} = 1$$

At s=-0.845

$$|k| = 3.079$$

When k=48 root locus intersects at jω axis

For the damped oscillatoryresponse roots on the locus should have complex values and in the left half of S-plane.

When 3.079<k<48 the roots will be complex and the system has damped oscillatory response

(iii) Highest value of K for continuous oscillations is the value of K when root locus crosses the imaginary axis i.e., K = 48

(iv) The frequency of oscillations at K = 48 is 2.828 rad/sec

(v) Value of K for a damping ratio of 0.8

Draw a line at an angle of 36.87°, with respect to negative real axis. This line intersects the root locus at s = -0.8+j0.6

Example : **Sketch the root locus for positive values of K for the system** $\boldsymbol{G(S)H(S) = \frac{k(s+6)}{s(s+1)(s+2)}}$ **the system is stable. Determine the K value at BAP.**

Sol.

l. Number of poles P = 3 at s=0,-1,-2.

2. Number of zeros Z = 1. At s=-6

3. Number of separate root loci = N = 3 (P > Z).

4. Number of asymptotes = P - Z = 3-1=2.

Angle of asymptotes,

$$\theta_k = \frac{(2k+1)}{P-Z}180° \ldots\ldots\ldots k = 0,1,$$

$$= \frac{(2k+1)}{3}180°$$

$$= 90°, 270°$$

5. Centroid $\sigma_1 = \frac{\sum p - \sum z}{P-Z}$

$$\sigma_1 = 1.5$$

6. Crossing of the root locus on jω axis

Characteristic eq. =$s^3 + 3s^2 + s(K+2)+ K=0$

From s^1 Row

$$3K + 6 - 6K = 0 \quad or\ k = 2$$

The value of K when root locus crosses the imaginary axis is 2

Routh array:

S^3	1	(K+2)
S^2	3	6K
S^1	(k+2)3-6K/6	
S^0	6K	

From s^2 Row

$3s^2+6K=0$

$S^2=-2$

And $\omega = \pm 2$

7. To determine Break away point (BAP)

From the characteristic equation, obtain expression for K

Characteristic equation 1 + G(s) H(s) = 0

$$K = \frac{-(s^3 + 3s^2 + 2s)}{(s+6)}$$

S=-0.4

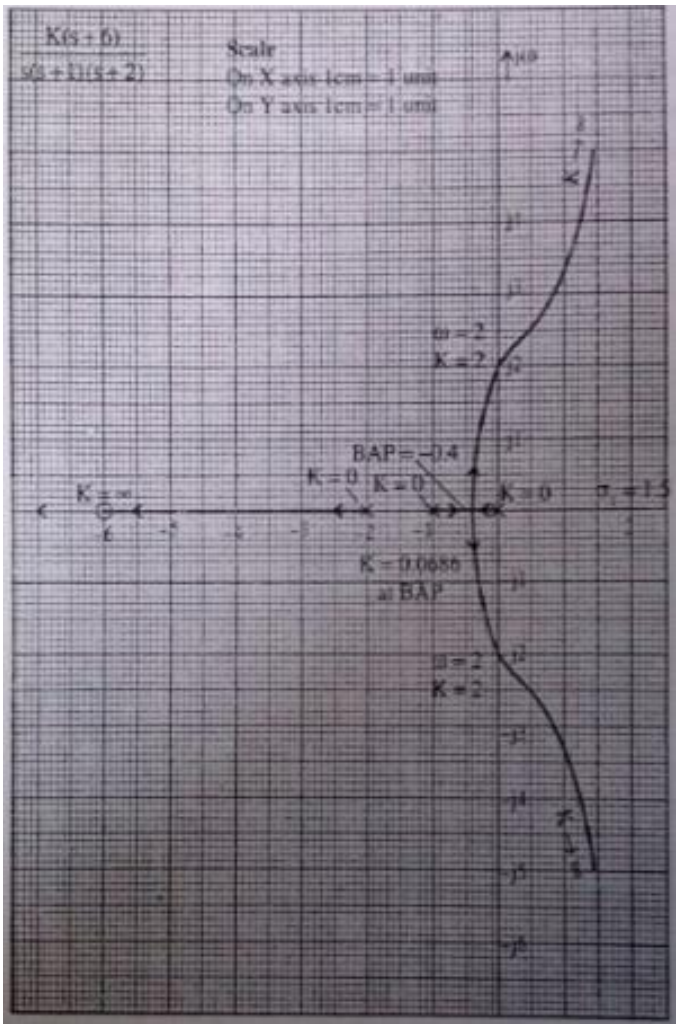
The root locus exist on real axis between 0 and -1 and between -2 and -6

Since the root locus exists between two neighbouring poles (s=0 and s=-1) there has to be one breakaway point in this section.

The system is stable when 0 < K < 2.When K > 2, root locus moves into right half of s-plane and hence the system will be unstable.

The value of 'K' at BAP :

The value of K at BAP (s = -0.4) is 0.0686



4.7 BODE PLOTS:

In order to represent sinusoidal transfer function of a system polar plots or Bode plots are commonly used. By plotting the frequency response of open loop transfer function of a system it is possible to determine the closed loop behaviour of the system. In case of a polar plot both Magnitude and phase angle are plotted in a single plot [It is a magnitude versus phase plot].

A Bode plot consists of two separate plots-one is a plot of magnitude of transfer function in decibels versus frequency and the other is a plot of phase angle of transfer function versus frequency. Usually graphs are plotted as a function of frequency ω in which case semi-log paper is used with logarithmic scale for frequency. Linear scale is

used for logarithmic magnitude of transfer function [i.e, 20 log |G(jw)|] and phase angle in degrees.

BASIC PRINCIPLE:

In general the open loop transfer function of a control system is of the form,

$$G(S)H(S) = \frac{K(1+T_aS)(1+T_bS)\ldots\ldots\ldots(1+T_mS)}{S^n(1+T_1S)(1+T_2S)\ldots\ldots\ldots[1+2\zeta\frac{S}{\omega_n}+\left(\frac{S}{\omega_n}\right)2]}\ldots\ldots.(1)$$

This form of transfer function is known as time constant form. For drawing Bode plots the given transfer function is to be written in time constant form. The sinusoidal transfer function is obtained by setting S= jω in above Equation.

$$G(\mathrm{j}\omega)H(v) = \frac{K(1+\mathrm{j}\omega T_a)(1+\mathrm{j}\omega T_b)\ldots\ldots\ldots(1+\mathrm{j}\omega T_m)}{(\mathrm{j}\omega)^n(1+\mathrm{j}\omega T_1)(1+\mathrm{j}\omega T_2)\ldots\ldots\ldots[1+2\zeta\frac{\mathrm{j}\omega}{\omega_n}-\left(\frac{\omega}{\omega_n}\right)2]}\ldots\ldots\ldots(2)$$

Magnitude plot:

In order to obtain the magnitude plot the procedure is as follows: Taking log magnitude of above Equation

$$20\log|G(\mathrm{j}\omega)H(\mathrm{j}\omega)| = 20\log\left|\frac{K}{(\mathrm{j}\omega)^N}\right| + 20\log|1+\mathrm{j}\omega T_a| + 20\log|1+\mathrm{j}\omega T_b| + \cdots -$$

$$20\log|1+\mathrm{j}\omega T_1| - 20\log|1+\mathrm{j}\omega T_2| - \cdots \ldots - 20\log|1+2\zeta\frac{\mathrm{j}\omega}{\omega_n} -$$

$$\left(\frac{\omega}{\omega_n}\right)2|$$

For plotting the magnitude plot, each term of above equation is considered and the magnitude plots of individual factors are plotted separately. The resultant magnitude plot of the transfer function is obtained by adding the magnitude plots of individual factors. The advantage of representing log magnitude of transfer function is that product and division factors in G(jω) H(jω) become addition and subtractions respectively.

Phase angle plot:

The phase angle of the transfer function is

$$G(j\omega)H(j\omega) = \tan^{-1}\omega T_a + \tan^{-1}\omega T_b + \ldots\ldots\ldots - N(90^\circ) \ldots\ldots$$

$$\ldots\ldots -\tan^{-1}\omega T_1 - \tan^{-1}\omega T_2 - \ldots\ldots\ldots -\tan^{-1}\left(\frac{2\zeta\frac{\omega}{\omega_n}}{1-\frac{\omega^2}{\omega_n^2}}\right) \ldots\ldots.(3)$$

For different frequencies ranging from a low value to high value, phase angle is calculated from above Equation by substituting for (ω). This phase angle is plotted as a function of (ω).

Plots of individual factors

The various factors in Equation (2) belong to one of the following forms:

(a) A constant gain, K

(b) Poles at the origin, $\frac{1}{j\omega}$; or Zeros at the origin, $\frac{j\omega}{1}$

(c) Poles on real axis, $\frac{1}{1+j\omega T}$ or Zeros on real axis, $\frac{1+j\omega T}{1}$

(d) Complex poles, $\frac{1}{1+j2\zeta\frac{\omega}{\omega n}-\left(\frac{\omega^2}{\omega n2}\right)}$ or complex zeros, $[1+j2\zeta\frac{\omega}{\omega_n}-\left(\frac{\omega^2}{\omega_n 2}\right)]$

The resonant frequency ω_r and the resonant peak value M_r

Consider a second order system with transfer function,

$$G(S) = \frac{1}{1+2\zeta\frac{S}{\omega_n}+\frac{S^2}{\omega_n{}^2}]} \ldots\ldots\ldots.(4)$$

The sinusoidal transfer function is,

$$G(j\omega) = \frac{1}{1+j2\zeta\frac{\omega}{\omega_n}-\frac{\omega^2}{\omega_n{}^2}]} \ldots\ldots.(5)$$

$$|G(j\omega)| = \frac{1}{\sqrt{(1-\frac{\omega^2}{\omega_n{}^2})^2+(2\zeta\frac{\omega}{\omega_n})^2}} \ldots\ldots.(6)$$

The magnitude of G($j\omega$) is a function of frequency and the magnitude has a peak value at a frequency known as resonant frequency.

To determine ω_r :

The magnitude of G($j\omega$) has a peak value when the denominator of Eqn. (6) has a minimum value.

$$\text{Let, } f(\omega) = (1 - \frac{\omega^2}{{\omega_n}^2})^2 + (2\zeta\frac{\omega}{\omega_n})^2 \ldots\ldots(7)$$

$$\frac{df(\omega)}{d\omega} = 2\left(1 - \frac{\omega^2}{{\omega_n}^2}\right)\left(-\frac{2\omega}{{\omega_n}^2}\right) + \frac{4\zeta^2(2\omega)}{{\omega_n}^2} \ldots\ldots\ldots(8)$$

At $\omega = \omega_r$, the expression $\frac{df(\omega)}{d\omega} = 0$

$$\frac{df(\omega_r)}{d\omega} = 2\left(1 - \frac{{\omega_r}^2}{{\omega_n}^2}\right)\left(-\frac{2\omega_r}{{\omega_n}^2}\right) + \frac{4\zeta^2(2\omega_r)}{{\omega_n}^2} = 0$$

$$\omega_r^2 = \omega_n^2(1 - 2\zeta^2)$$

The resonance peak occurs at

$$\omega_r = \omega_n\sqrt{1 - 2\zeta^2} \ldots\ldots(9)$$

The magnitude of $G(j\omega)$ at $\omega = \omega_r$

$$|G(j\omega_r)| = \frac{1}{\sqrt{(1 - \frac{{\omega_r}^2}{{\omega_n}^2})^2 + (2\zeta\frac{\omega_r}{\omega_n})^2}}$$

$$= \frac{1}{2\zeta\sqrt{1 - \zeta^2}} \ldots\ldots(10)$$

The resonance peak occurs if $\zeta < 0.707$

Steps to draw the Bode Plot:

(i) Express given G(s)H(s) into time constant form.
(ii) Draw a line of 20 Log K dB.
(iii) Draw a line of appropriate slope representing poles or zeros at the origin, passing through intersection point of $\omega = 1$ and 0 dB.
(iv) Shift this intersection point on 20 Log K line and draw parallel line to the line drawn in step 3. This is addition of constant K and number of poles or zeros at the origin.
(v) Change the slope of this line at various corner frequencies by appropriate value i.e. depending upon which factor is occurring at corner frequency. For a simple pole, slope must be changed by -20 dB / decade, for a simple zero by + 20 dB/ decade etc. Do not draw these individual lines. Change the slope of line obtained in step 5 by respective value and draw line with resultant slope. Continue this line

till it intersects next corner frequency line. Change the slope and continue. Apply necessary correction for quadratic factor.

(vi) Prepare the phase angle table and obtain the table of ω and resultant phase angle by actual calculation. Plot these points and draw the smooth curve obtaining the necessary phase angle plot.

NOTE: Remember that at every corner frequency slope of resultant line must change.

Example: **Draw the Bode magnitude (asymptotic) plot and phase angle plot for the open loop transfer function** $\boldsymbol{G(S)H(S) = \frac{48(S+10)}{S(S+20)(S^2+2.4S+16)}}$

Solution:

1. Rewrite the transfer function in time constant form.

$$G(S)H(S) = \frac{1.5\left(1+\frac{S}{10}\right)}{S\left(1+\frac{S}{20}\right)\left(1+\frac{2.4S}{16}+\frac{S^2}{16}\right)}$$

2. Obtain the sinusoidal TF

$$G(j\omega)H(j\omega) = \frac{1.5\left(1+\frac{j\omega}{10}\right)}{j\omega\left(1+\frac{j\omega}{20}\right)\left(1+\frac{2.4j\omega}{16}-\frac{\omega^2}{16}\right)}$$

3. List the factors and their corner frequencies:

	Factor	*Corner frequency*
(a)	Constant gain, K = 1.5	None
(b)	$\frac{1}{j\omega}$	None
(c)	$\left(1+\frac{j\omega}{10}\right)$	$\omega_c = 10$
(d)	$\frac{1}{1+\frac{j\omega}{20}}$	$\omega_c = 20$
(e)	$\frac{1}{\left(1+\frac{j2.4\omega}{16}-\frac{\omega^2}{16}\right)}$	$\omega_c = 4$

$[\omega_n^2 = 16, \therefore \quad \omega_n = 4]$

4. Arrange the factors in a table in ascending order of corner frequencies.

Factor	Corner frequency, ω_c	Asymptotic log magnitude characteristic	Resultant Slope
1) $\frac{1.5}{j\omega}$	None	Straight line of slope = –20db/dec passing through 20log 1.5 = 3.52 db at ω = 1. It passes through 0db at ω = 1.5	–20db/dec
2) $\frac{1}{\left(1+j\frac{2.4\omega}{16}-\frac{\omega^2}{16}\right)}$	$\omega_1 = 4$	Straight line of 0db for $\omega < \omega_1$. Straight line of slope –40db/dec for $\omega > \omega_1$	–60db/dec
3) $\left(1+\frac{j\omega}{10}\right)$	$\omega_2 = 10$	Straight line of 0db for $\omega < \omega_2$. Straight line of slope 20db/dec for $\omega > \omega_2$.	–40db/dec
4) $\frac{1}{1+\frac{j\omega}{20}}$	$\omega_3 = 20$	Straight line of 0db for $\omega < \omega_3$. Straight line of slope –20db/dec for $\omega > \omega_3$.	–60db/dec

5. Procedure to draw Asymptotic log magnitude plot:
   (a) At frequencies less than ω_r, only the factor $\frac{1.5}{j\omega}$ is effective. All other factors have a zero value. Therefore the slope of the resultant plot for $\omega < \omega_r$is -20 dB/dec. At $\omega = 1$ the resultant plot has avalue 3.52dB and at $\omega = 1.5$ the resultant plot has a value 0dB.
   (b) Above ω_1, the factor $\frac{1}{1+\frac{2.4j\omega}{16}-\frac{\omega^2}{16}}$has a slope of 40dB/dec and this slope is added to the previous slope to get resultant slope of the plot between ω_1 and ω_2. The resultant slope is 20 +(-40)=-60dB/dec.
   (c) Above ω_2, the factor $\frac{1}{1+\frac{j\omega}{10}}$ has a slope of +20dB/dec and this slope is added to the previous slope (i.e., -60dB/dec) to get resultant slope of the plot between ω_2 and ω_3. The resultant slope is -60+20= -40dB/dec.
   (d) Above ω_3, the factor $\frac{1}{1+\frac{j\omega}{20}}$ has a slope of -20dB/dec and this slope is added to the previous slope (i.e.,-40dB/dec) to get resultant slope of the plot above ω_3. The resultant slope is -40+(-20)=-60db/dec.

The asymptotic log magnitude curve is drawn on semilog sheet. The starting frequency on frequency scale should be less than lowest comer frequency (i. e., ω_1). If $\omega_0$is the starting frequency then the magnitude at $\omega_0$is calculated by considering $|\frac{K}{j\omega}|$ factor.

Here ω_0 is selected as 0.1 rad/sec.

Therefore magnitude at ω_0 is 23.5 dB

6. Phase angle curve

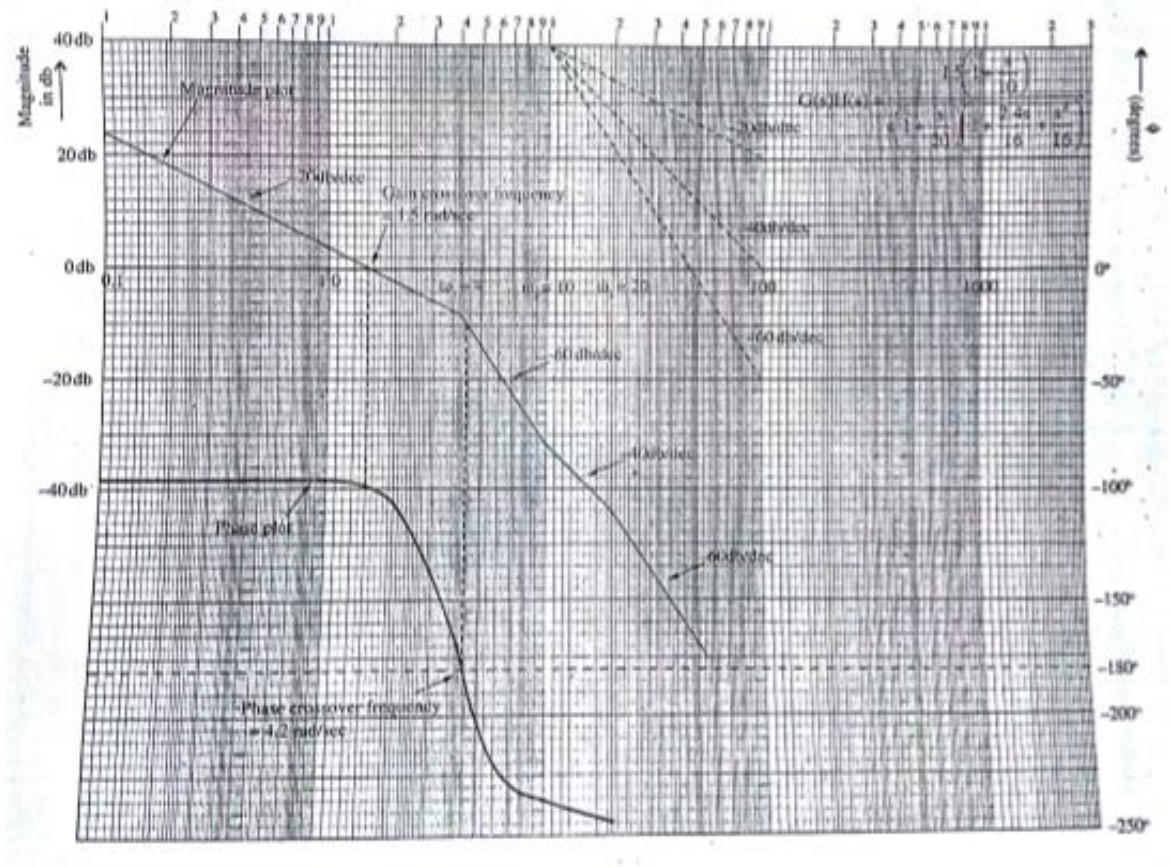
The phase angle of the sinusoidal transfer function is given by,

$$\angle G(j\omega)H(j\omega) = \tan^{-1}\frac{\omega}{10} - 90° - \tan^{-1}\frac{\omega}{20} - \tan^{-1}\frac{\frac{2.4\omega}{16}}{1-\frac{\omega^2}{16}}$$

Phase angle is calculated at various frequencies

ω	G(jω)H(jω)
1	–96.24°
2	–106.2°
4	–169.5°
5	–204.3°
8	–233°
10	–235.6°
20	–244.4°

By using the above data, phase angle curve is drawn.



To determine gain margin and phase margin

The crossing of zero dB line by magnitude plot is known as gain crossover frequency, ω_g..From the plot, $\omega_g = 1.5 rad/sec$.

At ω_g draw a perpendicular line which intersects the phase plot and obtain the phase angle at ω_g. From the plot it is –l00°.

Phase margin (PM)=180° + phase angle at gain cross over frequency

= 180°-100°=80°

The frequency at which phase angle is -180° is known as phase crossover frequency,ω_p.From the plot, $\omega_p = 4.2\ rad/sec$.

At this point draw a perpendicular line which intersects the magnitude plot. This point of intersection gives the magnitude at phase crossover frequency. From the plot it is -10dB.

Gain margin (GM) = -[Magnitude at phase crossover frequency]

= -(-10)

=10dB

Unit – 5

Introduction to Design and Advances in Control System

5.1 The Design Problem

The design of control systems is a demonstration of the art of compromise: a system is made a millisecond slower but a millimeter more accurate using an ounce less torque. A designer is often given the task of designing a controller that will allow the closed-loop system to meet some dynamic performance requirements without violating certain constraints. It is the interplay of performance requirement against performance